Dataset summary
MLP baseline
Macro-F1 focus

# The Pokemon are out there! 2026
# Task 1

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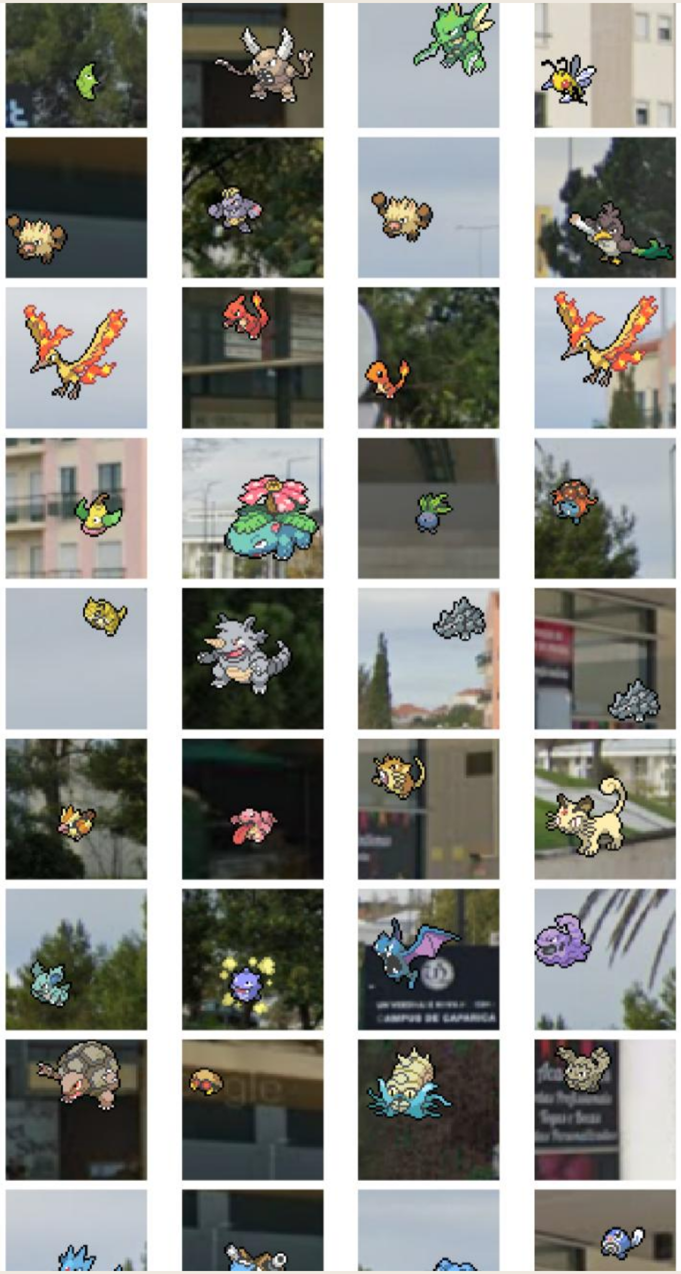
3,600 train images • 9 classes • Best validation macro-F1: 0.2132

Assignment 1 review

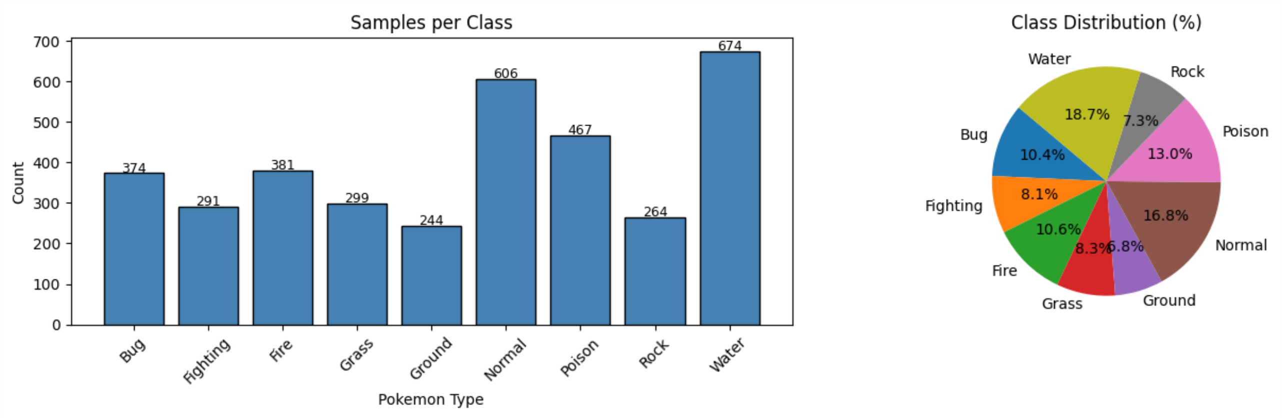


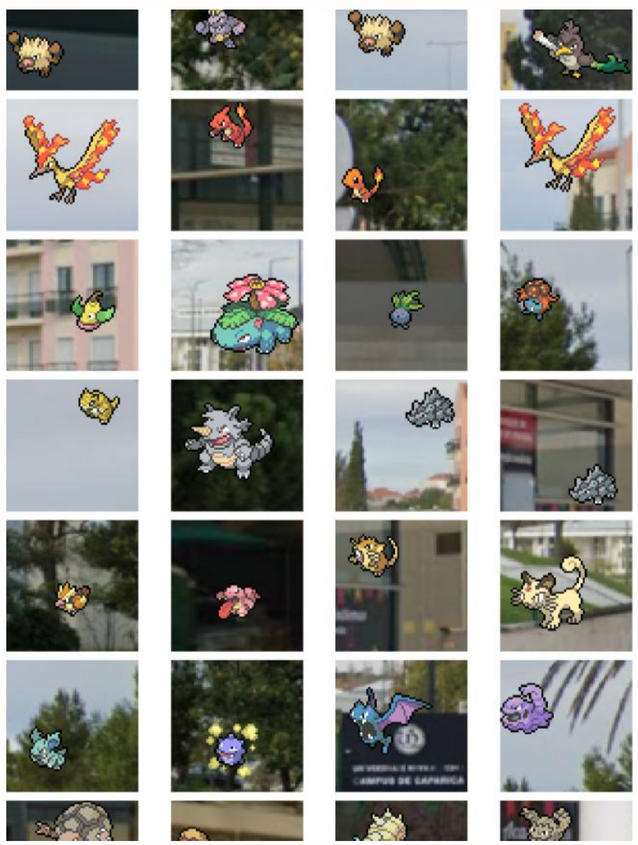
# 1. Data exploration insights



The dataset is manageable but not balanced, so class-sensitive evaluation is more informative than accuracy alone.

- 3,600 labeled training images spread across 9 Pokémon types.
- Each image is 64×64, which limits fine-grained texture and shape detail.
- Label frequencies vary from 244 (Ground) to 674 (Water): a 2.8× imbalance.



Sample sprites show why some classes are easier: Fire/Water are visually distinctive, while Rock/Ground/Fighting often share earthy colors and similar silhouettes.

# 2. Model architecture and rationale

Simple baseline, heavy regularization

| Input | Flatten | Dense | Dense | Dense | Dense | Logits |
|---|---|---|---|---|---|---|
| 96×96×3 | 27,648 | 2048 | 1024 | 512 | 256 | 9 classes |

BatchNorm + ReLU after each hidden layer
Dropout = 0.4 on early layers, 0.3 before the output

## Why this design

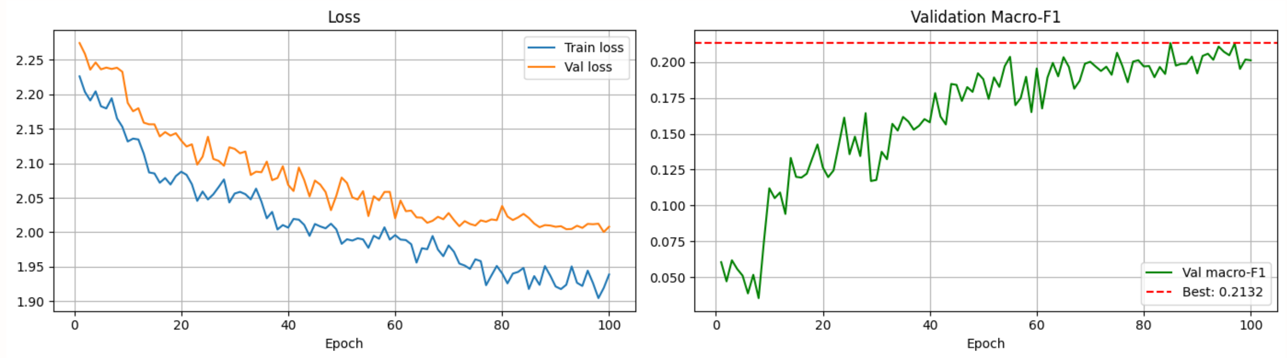
- Uses aggressive augmentation before flattening to expand the small training set.
- WeightedRandomSampler and class-weighted cross-entropy both counter label imbalance.
- Adam + ReduceLROnPlateau make optimization stable for a large fully connected network.
- Trade-off: flattening discards spatial locality, while the model still carries 59.4M trainable parameters.

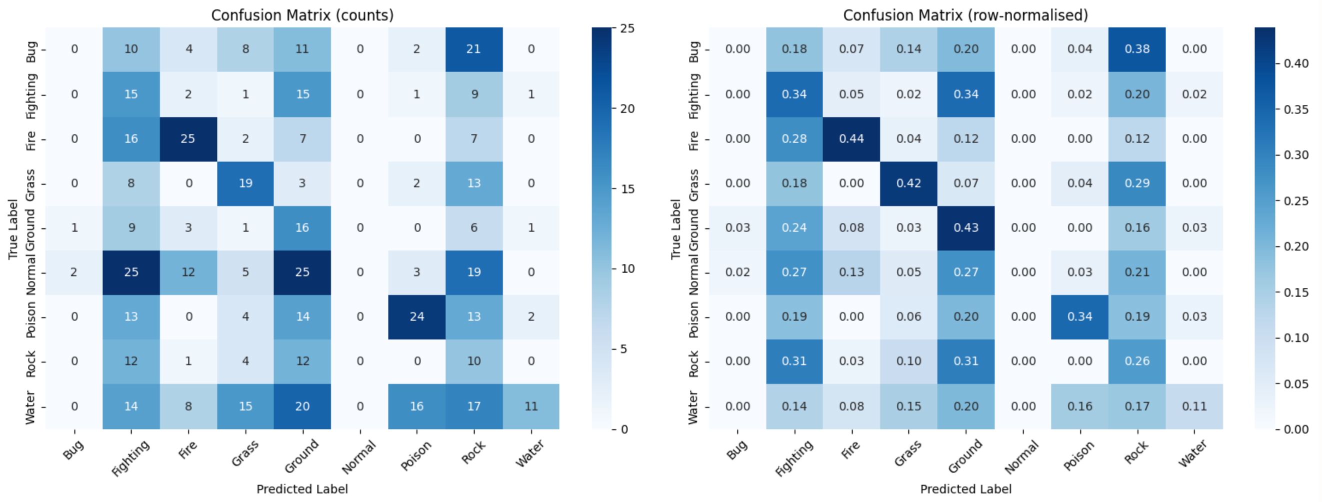
### Training setup

- 15% stratified validation split
- Batch size 64
- Learning rate 3e-4
- Up to 100 epochs, patience 20

# 3. Performance metrics and interpretation






## Headline metrics

Best validation macro-F1: 0.2132
Accuracy: 0.22
Macro precision / recall: 0.26 / 0.26

## Class-level pattern

Strongest F1: Fire 0.45, Poison 0.41, Grass 0.37
Weakest F1: Bug 0.00, Normal 0.00, Rock 0.13
Water precision is high (0.73) but recall is very low (0.11).

## Interpretation

- Validation loss keeps falling, but macro-F1 levels off near 0.20.
- Biggest confusion cluster: Ground, Rock, Fighting, and Grass.
- The baseline separates obvious classes, but flattening limits spatial feature learning that a CNN would preserve.